

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

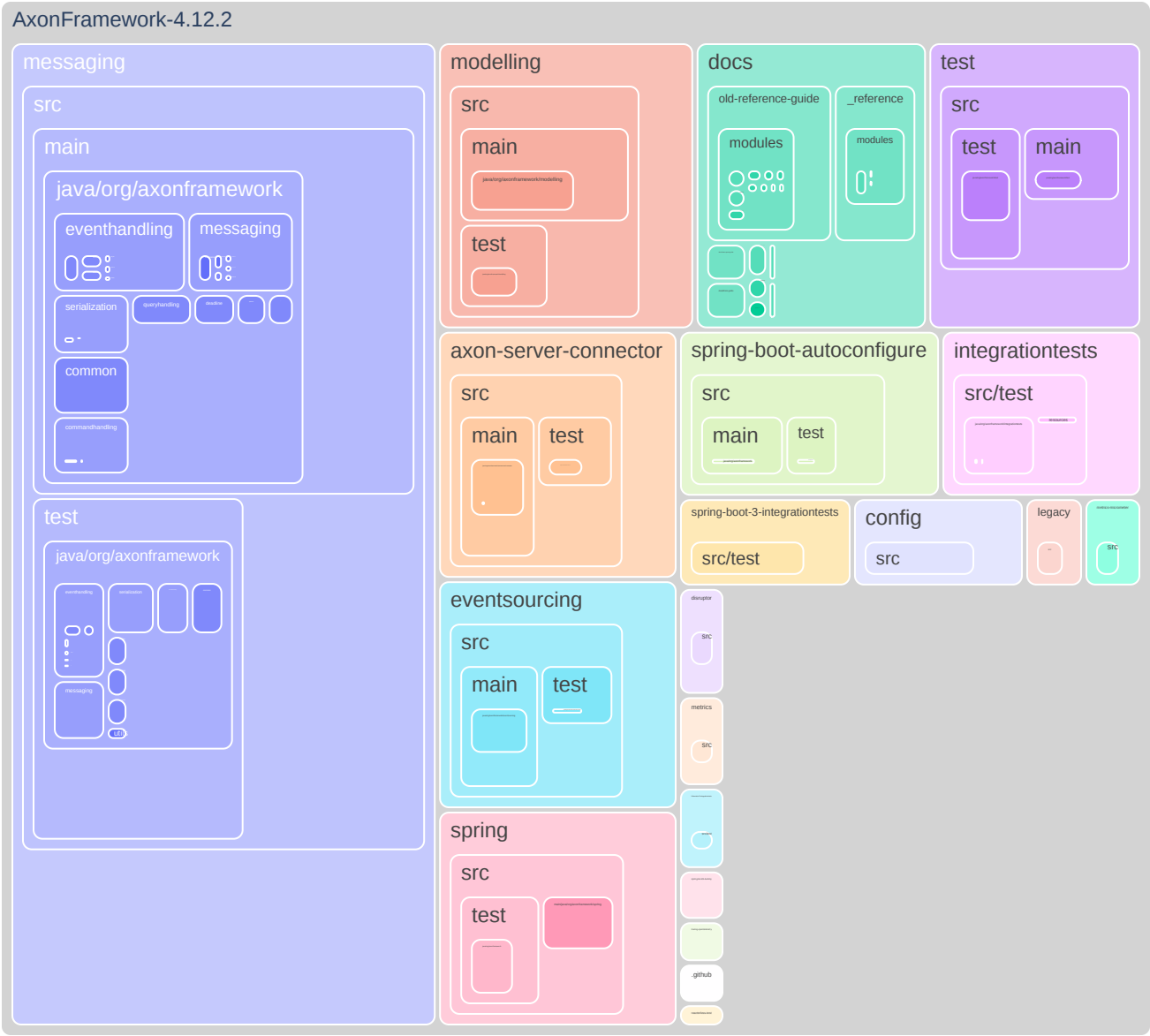
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	566.000000	566.000000	566.000000	566.000000	566.000000	566.000000
mean	24.867491	19.606007	288.137809	374.904594	1259.436396	469.128975
std	120.564946	21.286048	640.533206	187.648018	982.306000	402.675621
min	1.000000	2.000000	3.000000	19.000000	24.000000	19.000000
25%	1.250000	7.000000	27.000000	249.000000	523.000000	311.000000
50%	4.000000	12.000000	81.500000	482.000000	1042.000000	482.000000
75%	10.000000	25.000000	269.250000	482.000000	2058.500000	482.000000
max	2295.000000	213.000000	9840.000000	2632.000000	5760.000000	3111.000000

	directoryPath	directoryParentPath
0	AxonFramework-4.12.2/assembly	AxonFramework-4.12.
1	AxonFramework-4.12.2/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat/source	AxonFramework-4.12.2/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/heartbeat
2	AxonFramework-4.12.2/axon-server-connector/src/main/resources/META-INF/services	AxonFramework-4.12.2/axon-server-connector/src/main/resources
3	AxonFramework-4.12.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event/util	AxonFramework-4.12.2/axon-server-connector/src/test/java/org/axonframework/axonserver/connector/event
4	AxonFramework-4.12.2/config/src/test/resources/META-INF	AxonFramework-4.12.2/config/src/test/resources
5	AxonFramework-4.12.2/coverage-report	AxonFramework-4.12.
6	AxonFramework-4.12.2/docs/_playbook/localLinks	AxonFramework-4.12.2/docs/_playbook
7	AxonFramework-4.12.2/docs/_reference/modules/ROOT/attachments	AxonFramework-4.12.2/docs/_reference/modules/ROOT
8	AxonFramework-4.12.2/docs/_reference/modules/ROOT/partials	AxonFramework-4.12.2/docs/_reference/modules/ROOT
9	AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector
10	AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector/partials	AxonFramework-4.12.2/docs/_reference/modules/axon-server-connector
11	AxonFramework-4.12.2/docs/_reference/modules/configuration/pages	AxonFramework-4.12.2/docs/_reference/modules/configuration
12	AxonFramework-4.12.2/docs/_reference/modules/configuration/partials	AxonFramework-4.12.2/docs/_reference/modules/configuration
13	AxonFramework-4.12.2/docs/_reference/modules/disruptor/pages	AxonFramework-4.12.2/docs/_reference/modules/disruptor
14	AxonFramework-4.12.2/docs/_reference/modules/disruptor/partials	AxonFramework-4.12.2/docs/_reference/modules/disruptor
15	AxonFramework-4.12.2/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.12.2/docs/_reference/modules/eventsourcing
16	AxonFramework-4.12.2/docs/_reference/modules/eventsourcing/partials	AxonFramework-4.12.2/docs/_reference/modules/eventsourcing
17	AxonFramework-4.12.2/docs/_reference/modules/legacy/pages	AxonFramework-4.12.2/docs/_reference/modules/legacy
18	AxonFramework-4.12.2/docs/_reference/modules/legacy/partials	AxonFramework-4.12.2/docs/_reference/modules/legacy
19	AxonFramework-4.12.2/docs/_reference/modules/metrics/pages	AxonFramework-4.12.2/docs/_reference/modules/metrics
20	AxonFramework-4.12.2/docs/_reference/modules/metrics/partials	AxonFramework-4.12.2/docs/_reference/modules/metrics
21	AxonFramework-4.12.2/docs/_reference/modules/micrometer/pages	AxonFramework-4.12.2/docs/_reference/modules/micrometer
22	AxonFramework-4.12.2/docs/_reference/modules/micrometer/partials	AxonFramework-4.12.2/docs/_reference/modules/micrometer
23	AxonFramework-4.12.2/docs/_reference/modules/modeling/partials	AxonFramework-4.12.2/docs/_reference/modules/modeling
24	AxonFramework-4.12.2/docs/_reference/modules/spring/pages	AxonFramework-4.12.2/docs/_reference/modules/spring
25	AxonFramework-4.12.2/docs/_reference/modules/spring/partials	AxonFramework-4.12.2/docs/_reference/modules/spring
26	AxonFramework-4.12.2/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.12.2/docs/_reference/modules/springboot-starter
27	AxonFramework-4.12.2/docs/_reference/modules/springboot-starter/partials	AxonFramework-4.12.2/docs/_reference/modules/springboot-starter
28	AxonFramework-4.12.2/docs/_reference/modules/test/pages	AxonFramework-4.12.2/docs/_reference/modules/test
29	AxonFramework-4.12.2/docs/_reference/modules/test/partials	AxonFramework-4.12.2/docs/_reference/modules/test

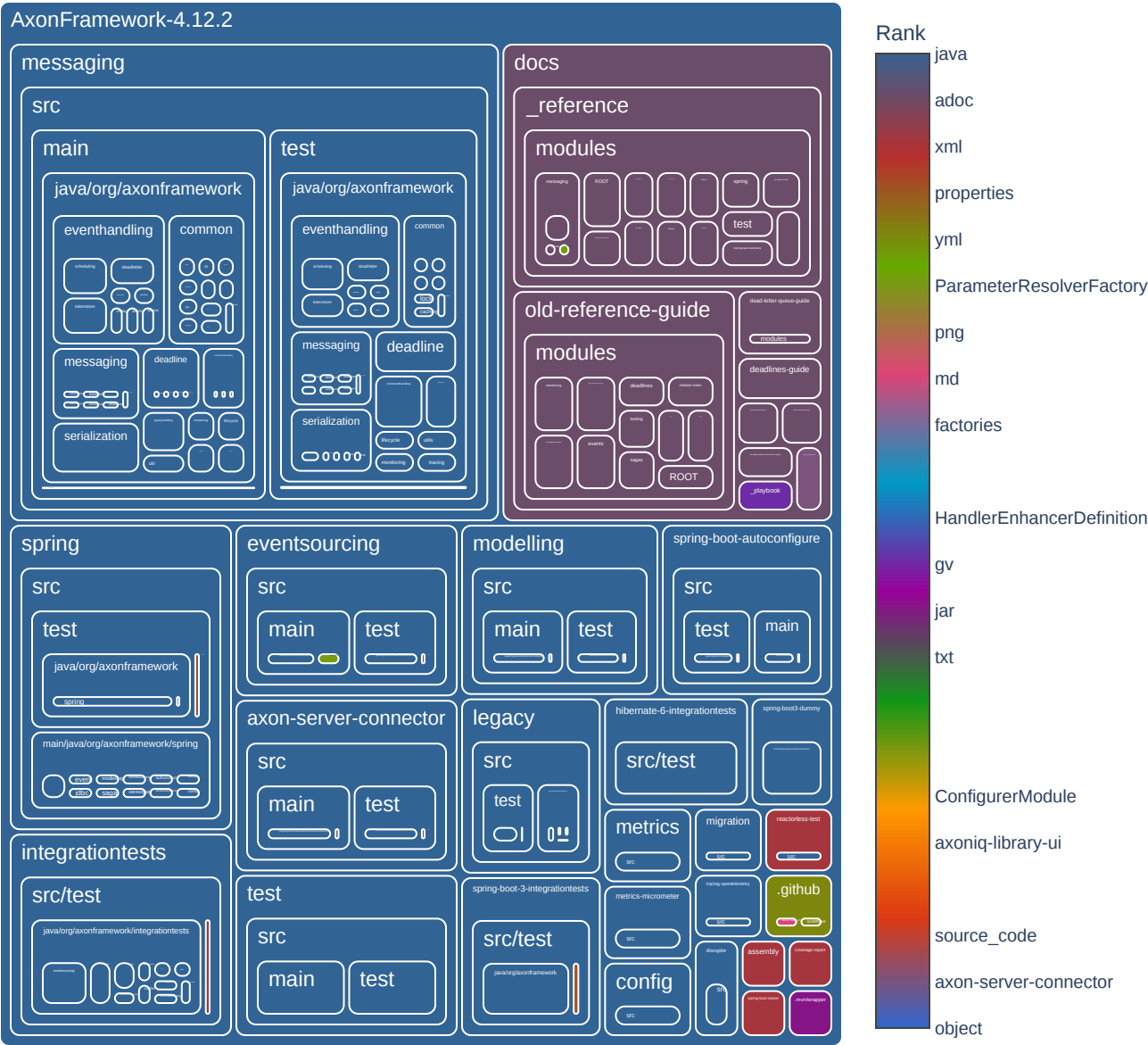
Number of files per directory

Directories and their file count



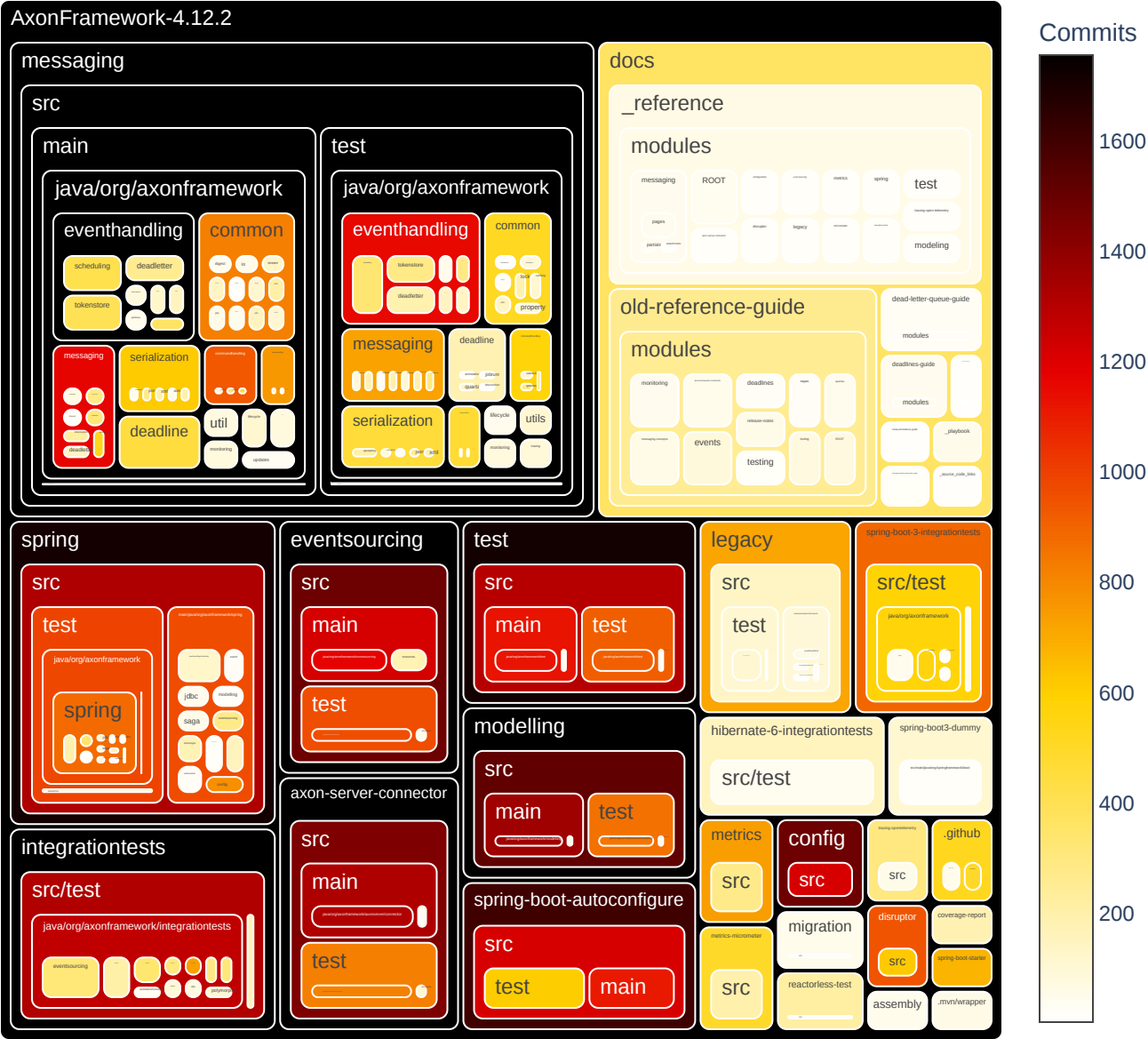
Most frequent file extension per directory

Most frequent file extension per directory



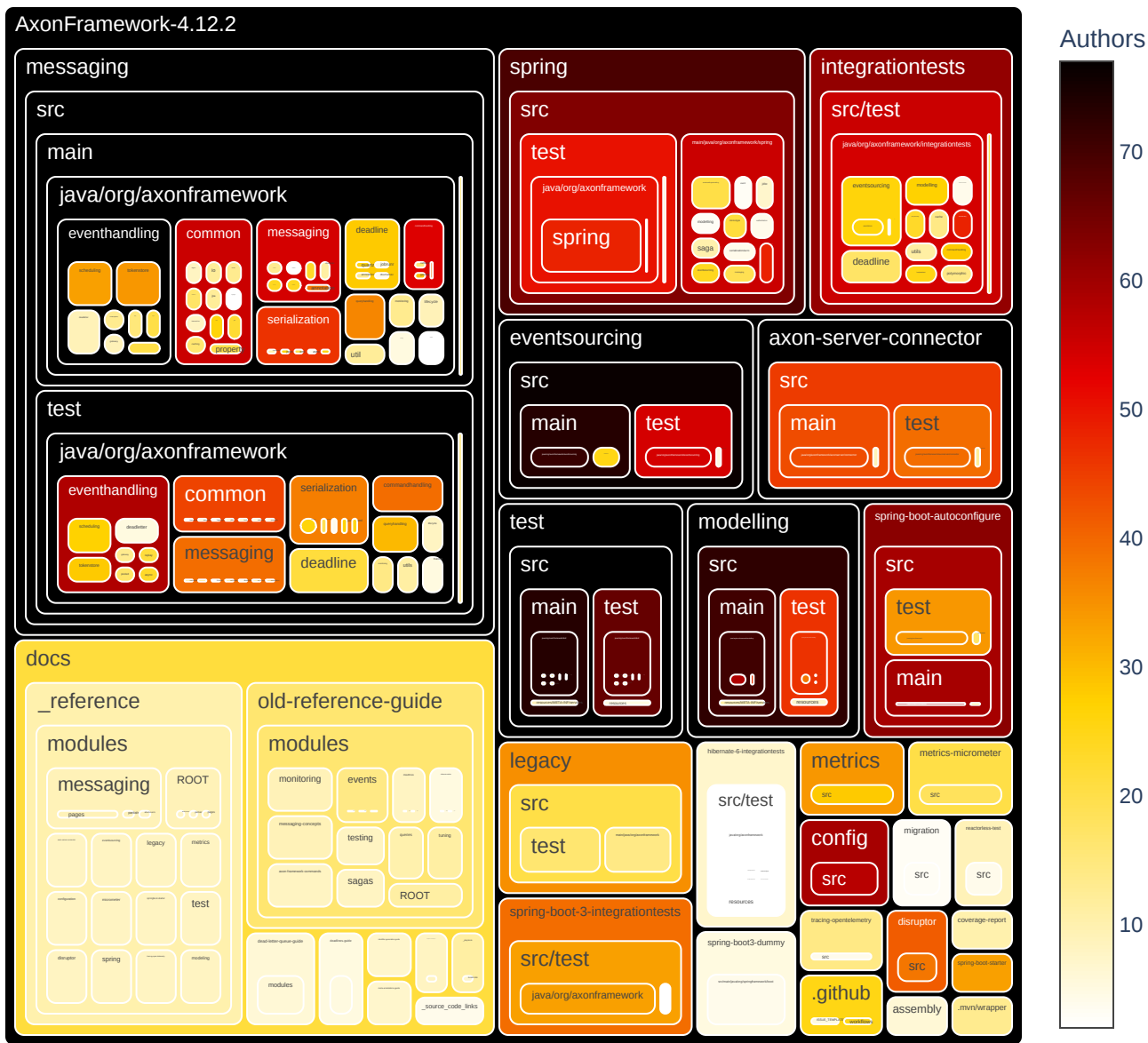
Number of commits per directory

Number of git commits



Number of distinct authors per directory

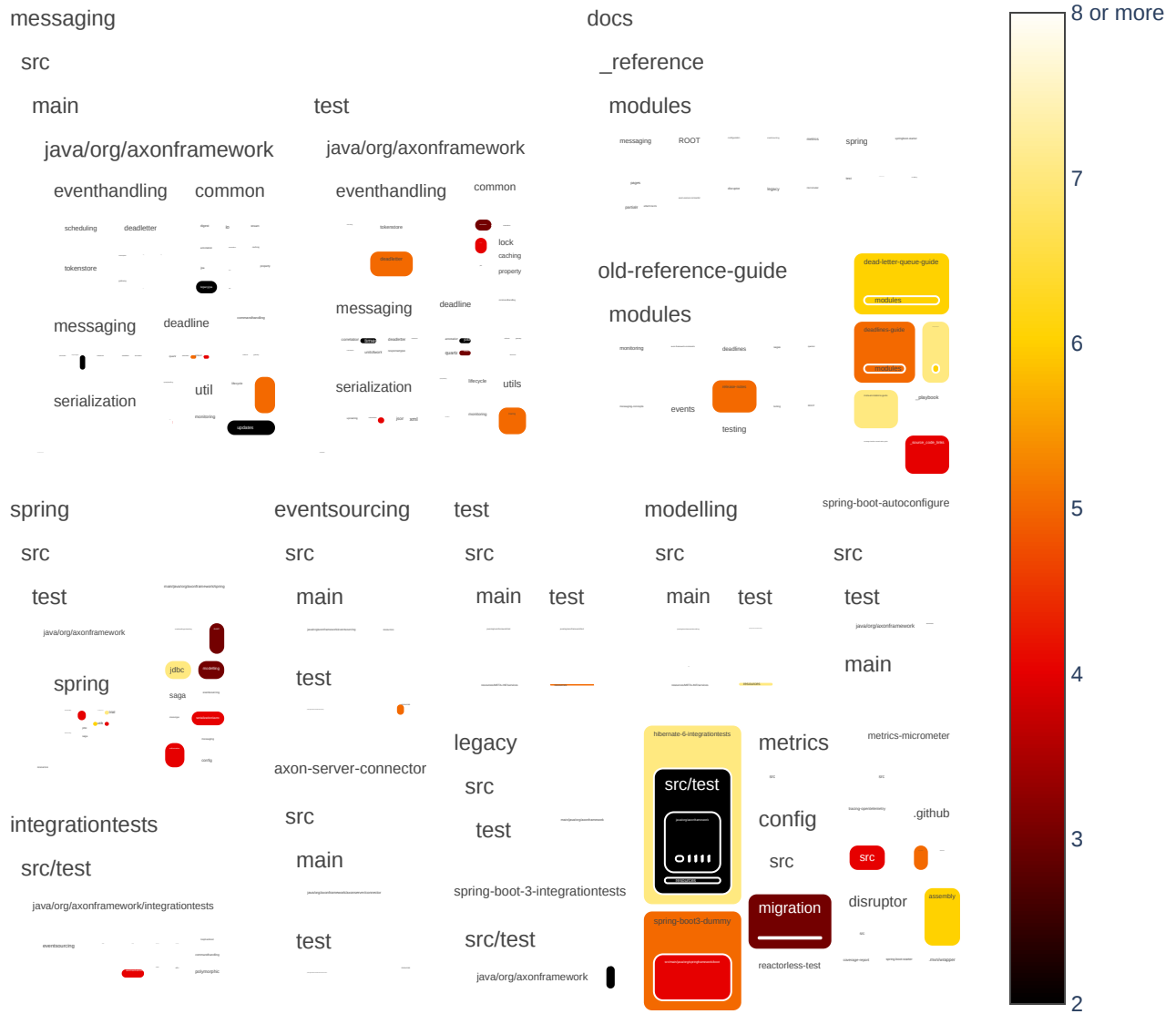
Number of distinct commit authors



Directories with very few different authors

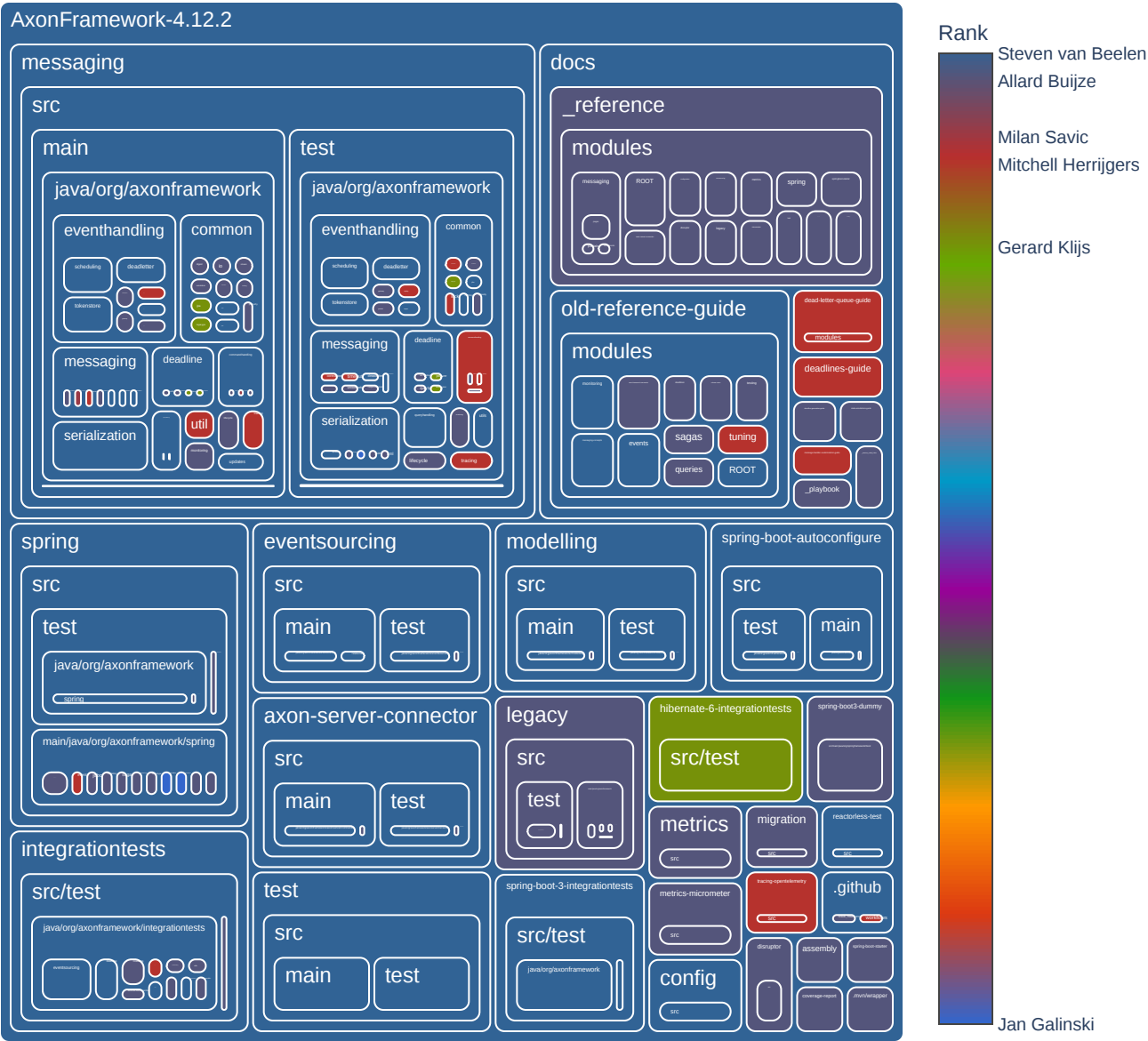
Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.12.2



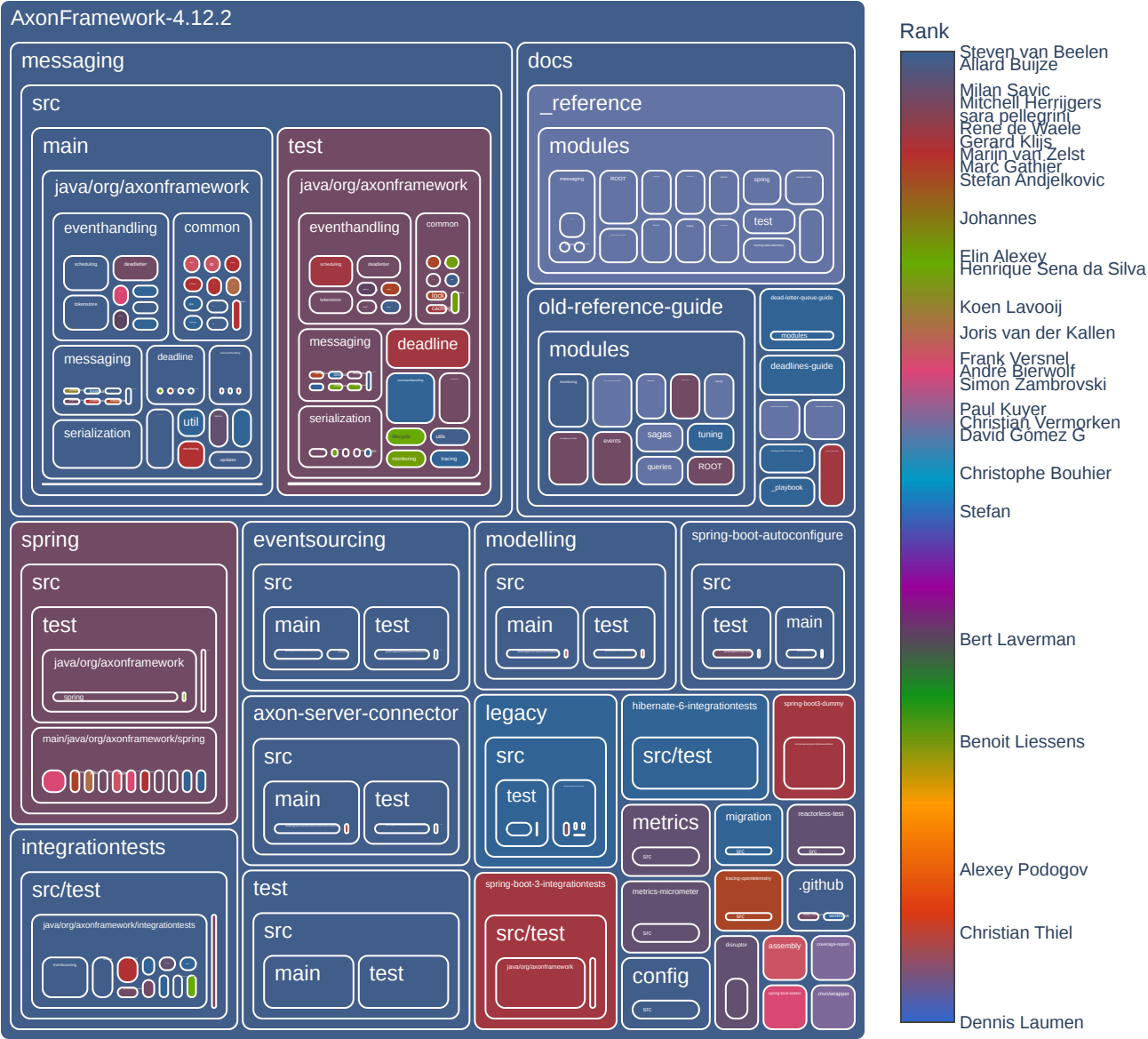
Main author per directory

Main authors with highest number of commits



Second author per directory

Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

AxonFramework-4.12.2

messaging

src

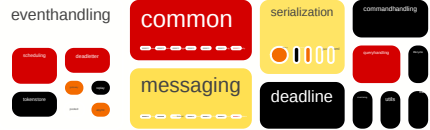
main

java/org/axonframework

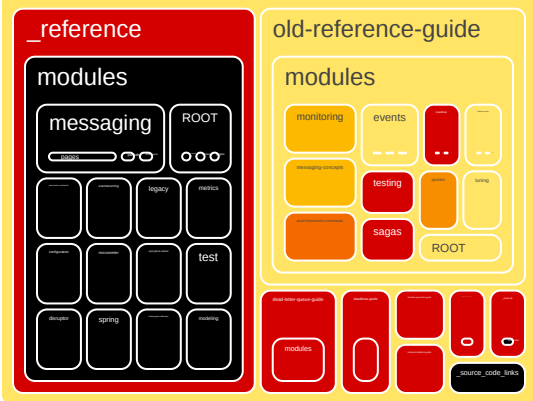


test

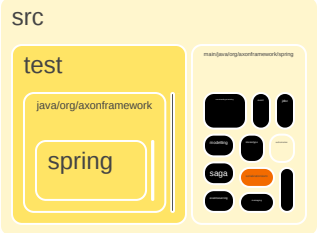
java/org/axonframework



docs



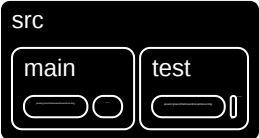
spring



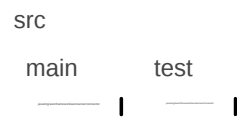
integrationtests



eventsourcing



axon-server-connector



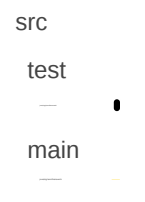
test



modelling



spring-boot-autoconfigure



legacy



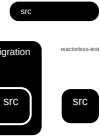
spring-boot-3-integrationtests



metrics



metrics-micrometer



spring-boot-3-integrationtests



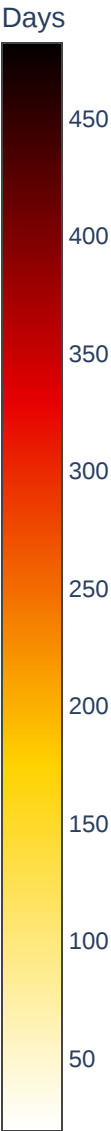
spring-boot-3-dummy



tracing-opentracing



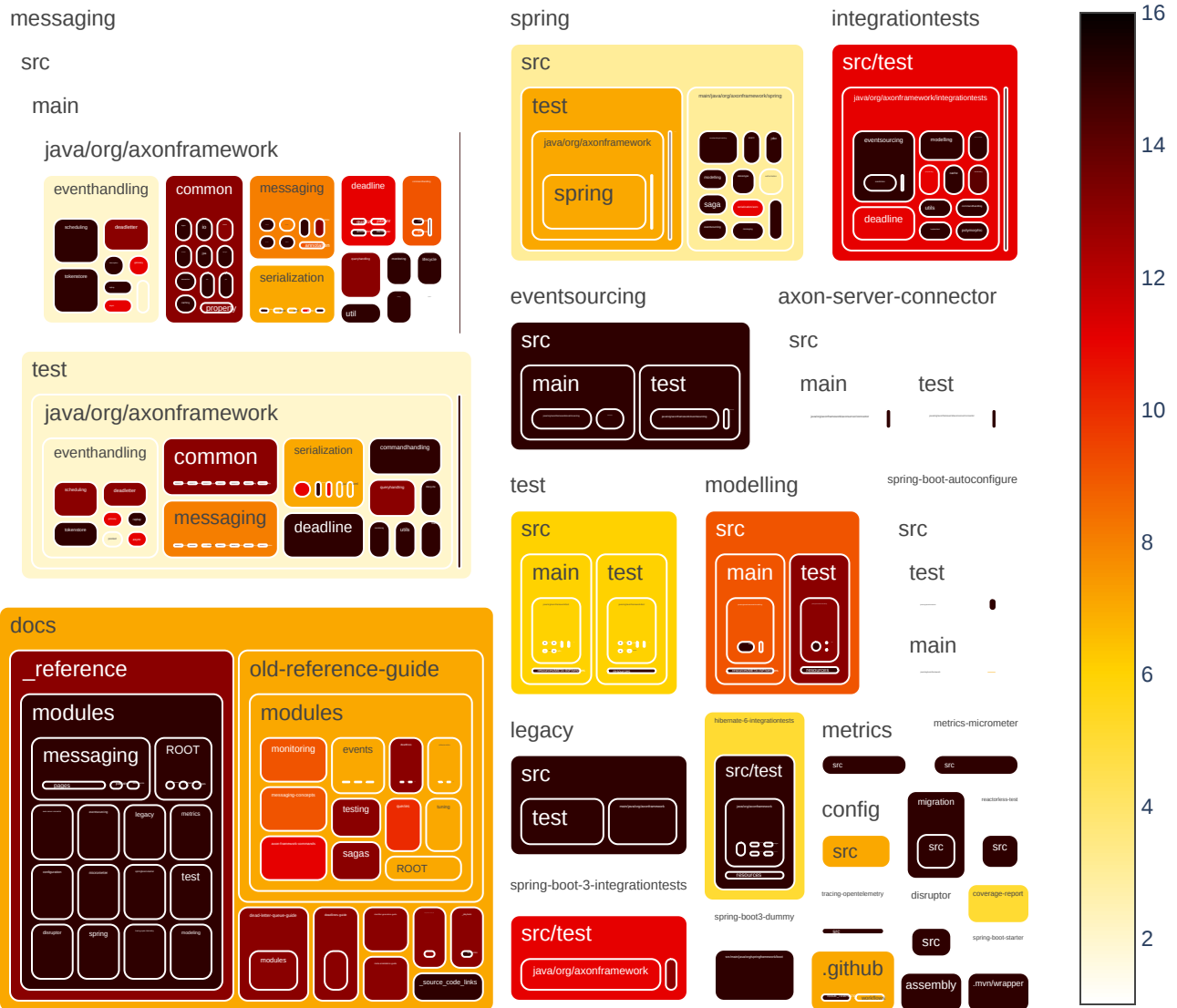
disruptor



Days since last commit per directory (ranked)

Rank of days since last commit

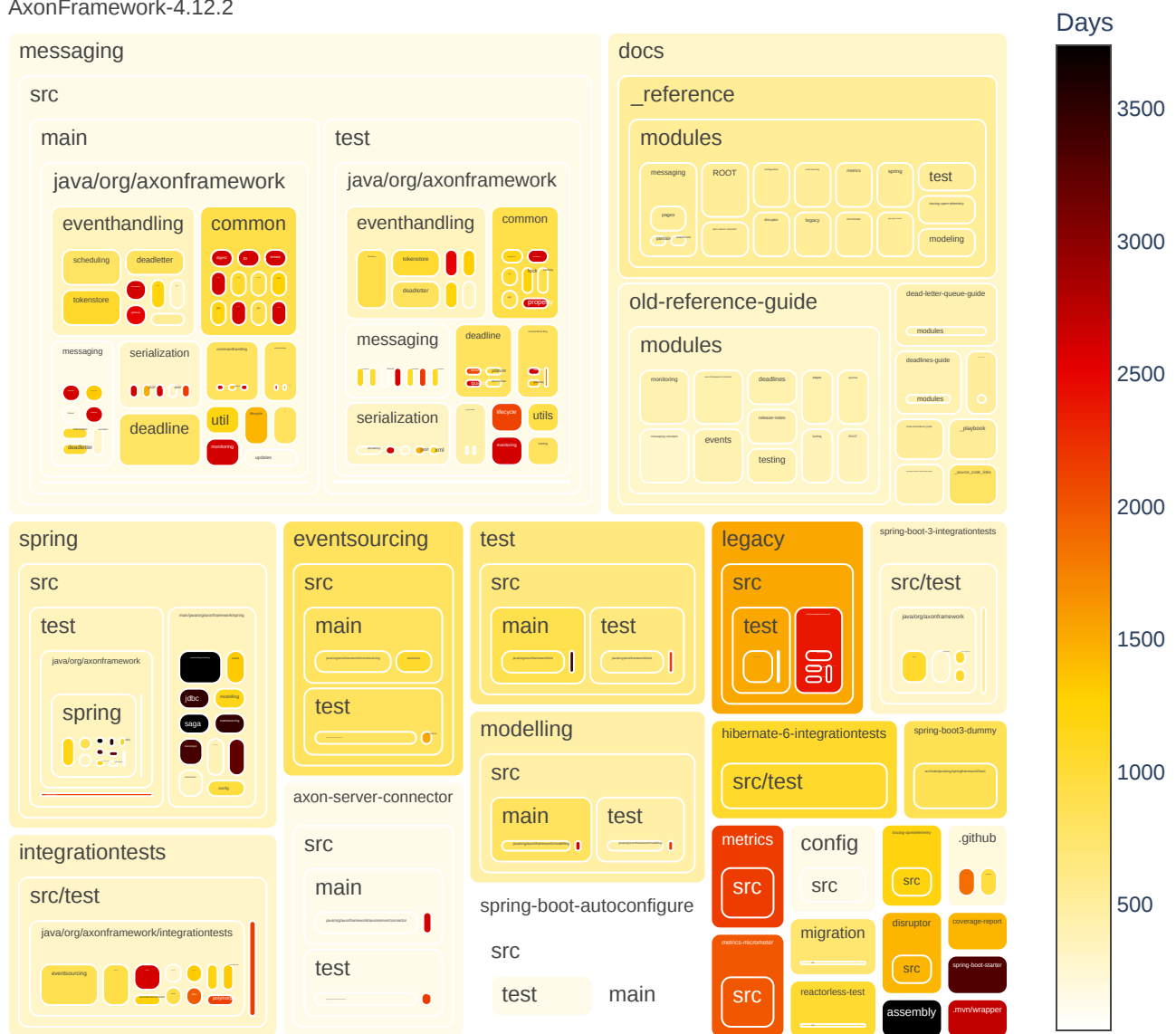
AxonFramework-4.12.2



Days since last file creation per directory

Days since last file creation

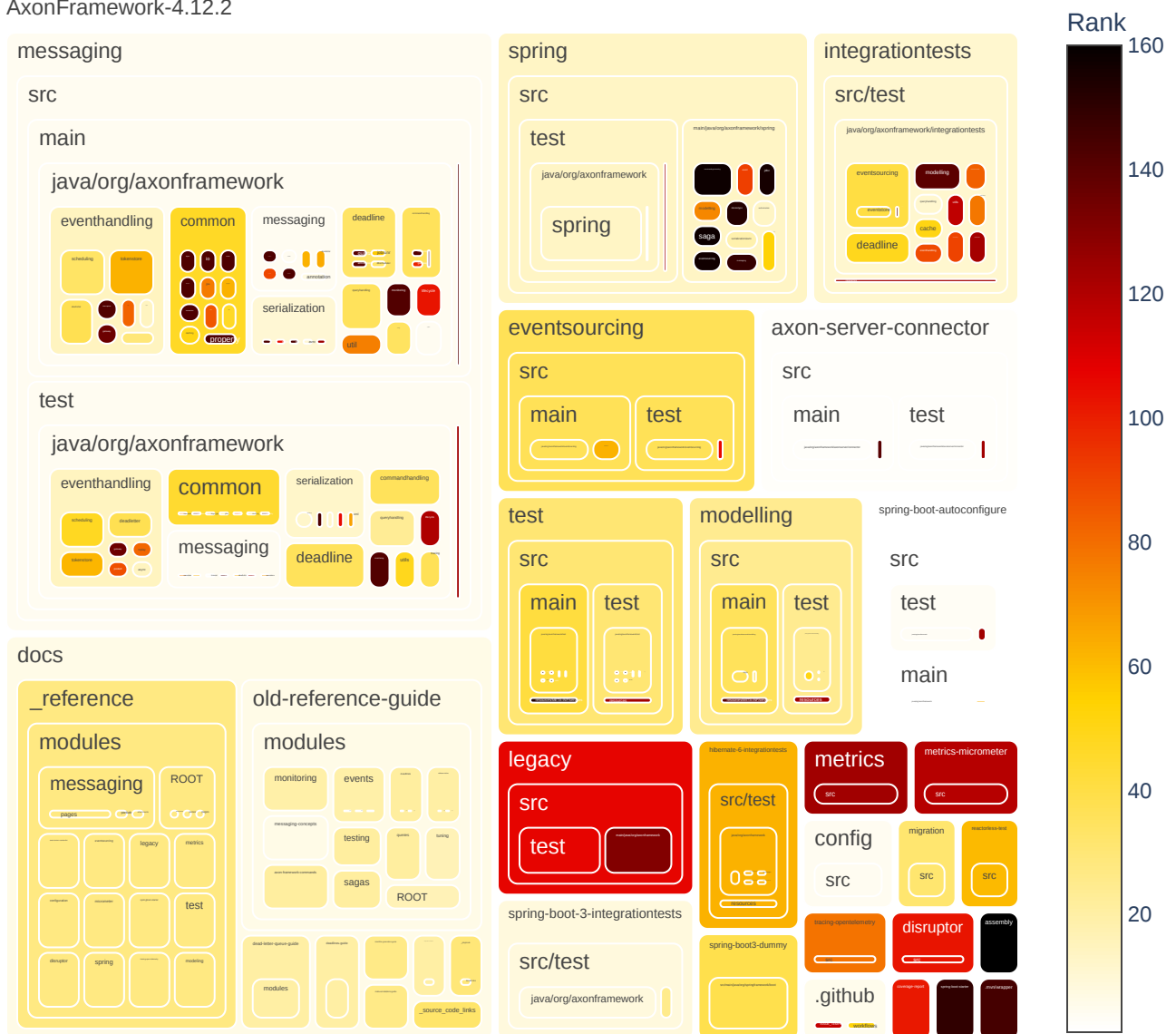
AxonFramework-4.12.2



Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.12.2



Days since last file modification per directory

Rank of days since last file modification

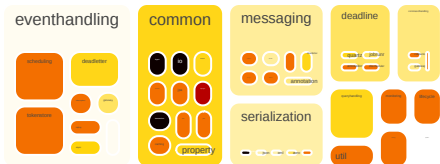
AxonFramework-4.12.2

messaging

src

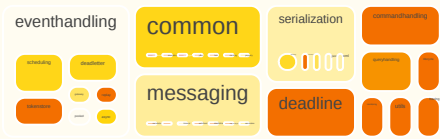
main

java/org/axonframework

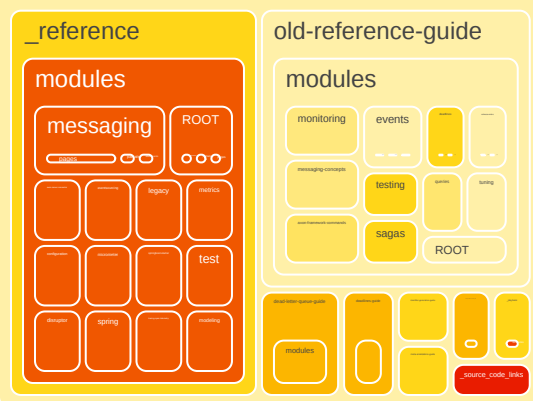


test

java/org/axonframework



docs



spring

src

test

java/org/axonframework

spring

eventsourcing

src

main

test

test

src

main

test

modelling

src

main

test

axon-server-connector

src

main

test

spring-boot-autoconfigure

src

test

main

legacy

src

test

spring-boot-3-integrationtests

src/test

java/org/axonframework

hibernate-6-integrationtests

src/test

0000

spring-boot3-dummy

0000

metrics

src

config

src

tracing-opentelemetry

0000

.github

assembly

metrics-micrometer

src

migration

src

disruptor

src

coverage-report

spring-boot-starter

assembly

min/wrappers

Rank

50

40

30

20

10

Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 973

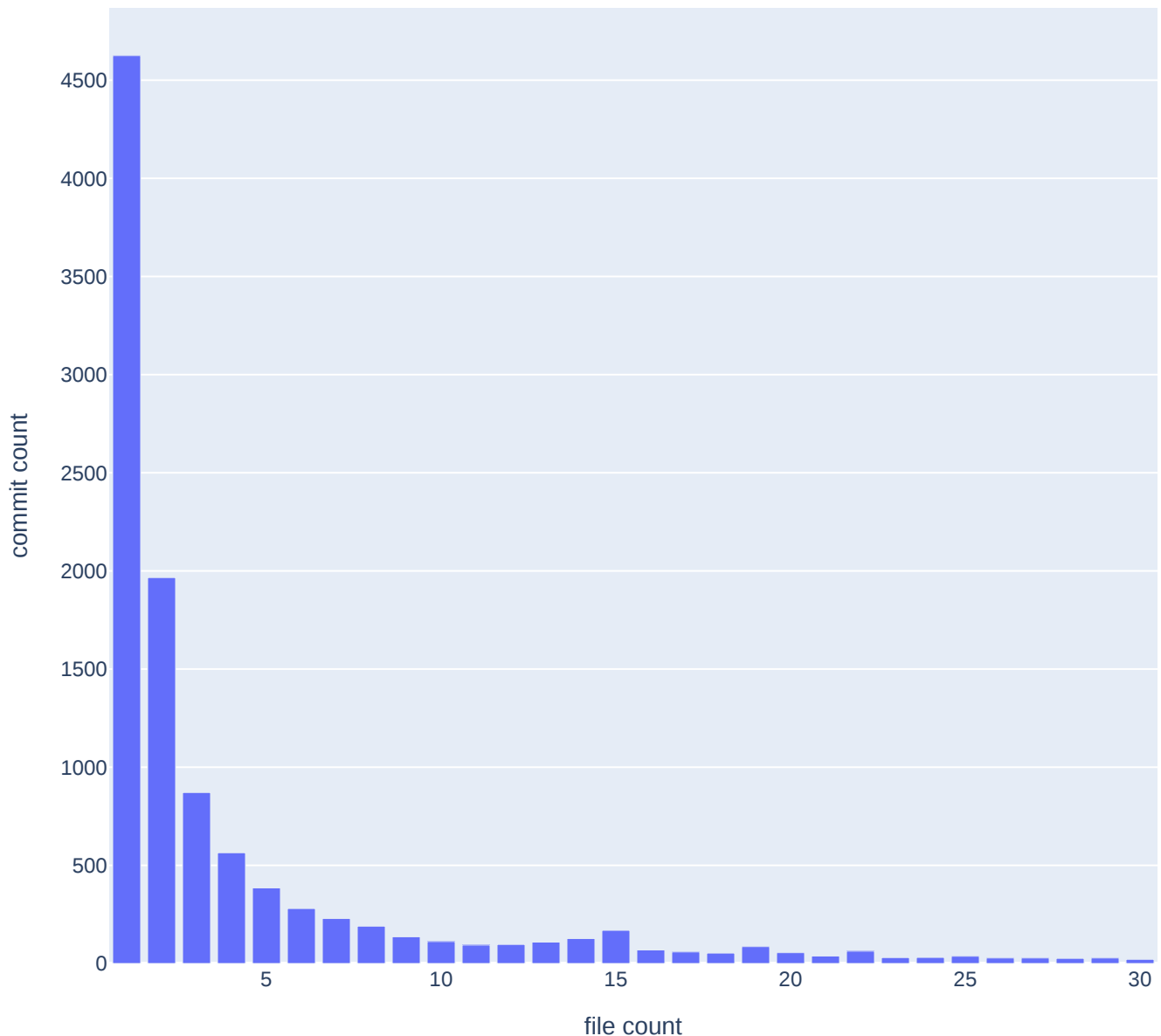
Max changed files with one commit = 2178

	filesPerCommit	commitCount
count	334.000000	334.000000
mean	343.449102	34.667665
std	439.862566	281.534044
min	1.000000	1.000000
25%	84.250000	1.000000
50%	185.000000	2.000000
75%	426.500000	5.000000
max	2178.000000	4625.000000

	filesPerCommit	commitCount
0	1	4625
1	2	1966
2	3	871
3	4	564
4	5	385
5	6	280
6	7	229
7	8	190
8	9	136
9	10	112
10	11	94
11	12	97
12	13	109
13	14	127
14	15	169
15	16	69
16	17	59
17	18	53
18	19	86
19	20	55
20	21	38
21	22	63
22	23	30
23	24	31
24	25	37
25	26	28
26	27	28
27	28	26
28	29	28
29	30	21

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files

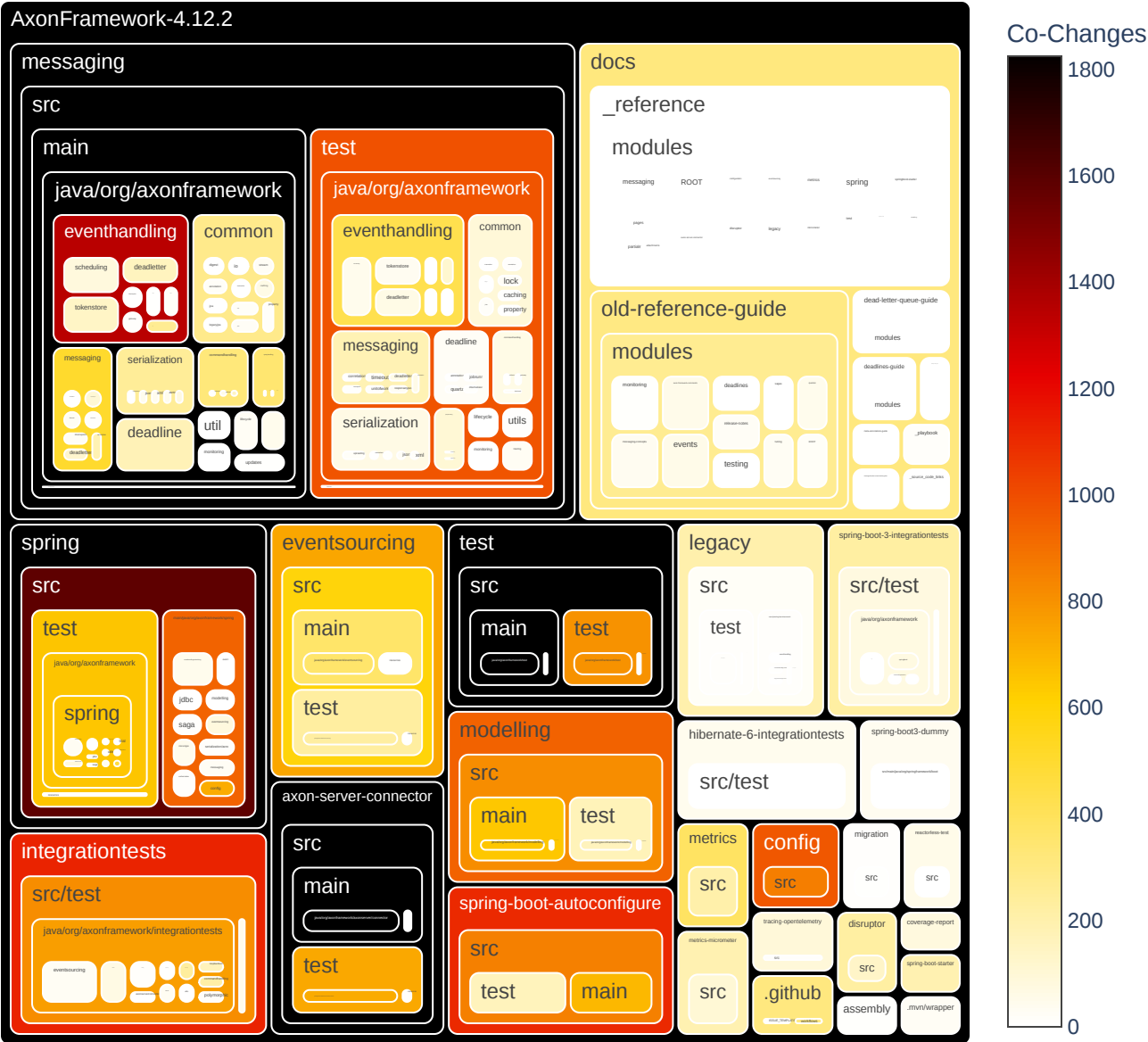
This section analyzes files that were changed together within the same commit and provides several metrics to quantify the strength of the co-change relationship:

- **Commit Count:** The number of commits in which two files were changed together.
- **Commit Lift:** A ratio that indicates whether the co-change pattern is stronger than random chance, given how often each file changes.
- **Jaccard Similarity:** The ratio of commits involving either file that also involved both files.

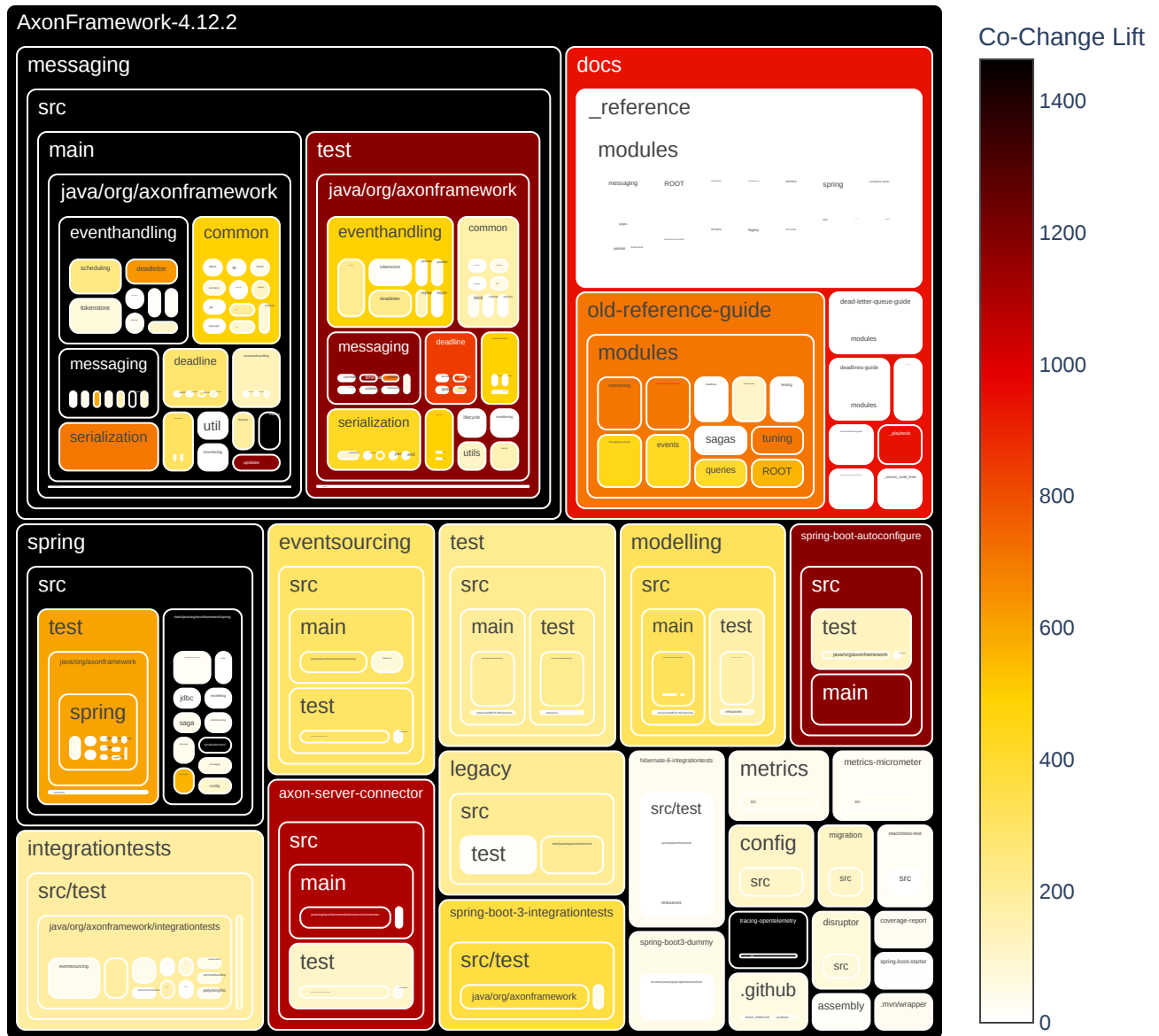
The following tables show the top pairwise changed files based on these metrics. The following charts show how these metrics are distributed across pairs of files that were changed together.

Treemap with files changed frequently with others

Files that likely co-change with others in update commits



Co-Changing files in update commits max lift (1=random, >1=more than random, <1=



Co-Changing files in update commits average lift (1=random, >1=more than random,

